

Disease-specific tau filaments assemble via polymorphic intermediates

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SITUATIONAL CONTEXT HOOK

This study investigates the in vitro assembly mechanism of recombinant human tau (residues 297–391) into filaments associated with Alzheimer’s disease (PHFs) and chronic traumatic encephalopathy (CTE). Using time-resolved cryo-EM, the authors identified a common, transient 'First Intermediate Amyloid' (FIA) filament, which forms as a shared intermediate at 120 minutes across different reaction conditions. The FIA core comprises residues 302–316, forming a hydrophobic interface that precedes the formation of disease-specific filament structures. The study demonstrates that subsequent maturation into diverse, polymorphic J-shaped and C-shaped protofilaments is influenced by reaction conditions (magnesium vs. sodium chloride), providing structural insights into primary and secondary nucleation processes in amyloid assembly. These findings challenge the requirement of non-filamentous oligomers as the primary drivers of amyloid formation in these models.

KEY FINDINGS (4)

Formation of a shared First Intermediate Amyloid (FIA) during the early stages of in vitro assembly of tau(297-391) in both PHF and CTE reaction conditions.

protein_pair Tau ↔ Tau
evidence 120 min (qualitative)
context Time-resolved cryo-EM analysis of tau assembly in vitro.
residues 302, 316
confidence 1.0 · significant: True
source_span Page 3, Para 2

The ordered core of the FIA consists of residues 302–316, where Val306 and Ile308 form a tightly packed hydrophobic interface.

protein_pair Tau ↔ Tau
evidence 1.7 to 3.8 Å resolution (quantitative)
context Atomic modelling based on cryo-EM density map.
residues 306, 308, 310
confidence 1.0 · significant: True
source_span Page 3, Para 3

Tau monomeric backbone displays a propensity to adopt extended conformations in residues 305–314 and 336–345, which are regions found in later β-strand structures.

evidence null (qualitative)
context Solution-state NMR of monomeric tau(297-391).
residues 305, 314, 336, 345
confidence 0.9 · significant: True
source_span Page 2, Para 1

Tau assembly involves maturation from J-shaped protofilaments to C-shaped protofilaments, mediated by rearrangements of 332PGGG335 and 364PGGG367 motifs.

evidence null (qualitative)
context Structural analysis of polymorphic intermediates across time points.
residues 332, 335, 364, 367
confidence 0.95 · significant: True
source_span Page 4, Para 1

CONTRADICTIONS AND NEGATIVE RESULTS (2)

FINDING	CONFLICTS WITH	REASONING
Thioflavin T fluorescence does not detect the FIA or many other early intermediate amyloid species.	False	Standard fluorescence-based monitoring is suboptimal for identifying early nucleated species in this system.
Structures of filaments from early assembly reactions often lack helical twist and are uninterpretable by standard reconstruction methods.	False	Transient, non-helical or heterogenous species at specific time points (e.g., 140-160 min in PHF reaction) prevent structure solution.

METHODOLOGY

Methods

- Time-resolved cryo-electron microscopy
- Solution-state NMR
- Analytical ultracentrifugation
- Thioflavin T fluorescence
- SDS-PAGE
- Helical reconstruction
- Atomic modelling

Instruments

- Thermo Fisher Krios G2
- Thermo Fisher Krios G4
- Falcon-4 camera
- Selectris X energy filter
- Bruker Avance III spectrometers
- Optima XL-I analytical ultracentrifuge
- Fluostar Omega

Controls

- Buffer-only reactions
- Time-zero sample collection
- Replicate reactions for reproducibility
- Internal FSC validation

Source organisms (NCBI taxon)

9606

PDB accessions (45)

8PP0 8Q27 8Q2J 8Q2K 8Q2L 8Q7F 8Q7L 8Q7M 8Q7P 8Q7T 8Q88 8Q8C 8Q8D 8Q8E 8Q8F 8Q8L 8Q8M 8Q8R 8Q8S 8Q8U 8Q8V 8Q8W 8Q8X 8Q8Y ...

ENTITIES

Proteins

tau, amyloid-β, α-synuclein, TDP-43

Chemicals

magnesium chloride, sodium chloride, thioflavin T, ammonium sulfate, IPTG

NORMALISED IDENTIFIERS (SIDECAR; 5 RESOLVED, 0 UNRESOLVED, 0 FILTERED)

RAW NAME	GENE	UNIPROT	TIER
Tau	MAPT	P10636	hgnc_alias
tau	MAPT	P10636	hgnc_alias
amyloid-β	APP	P05067	curated_alias
α-synuclein	SNCA	P37840	curated_alias
TDP-43	TARDBP	Q13148	hgnc_alias

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